Questions by topic: Habitat case studies: woodland, rainforest, reefs, islands, mangroves, Antarctica - Set 2 (60 marks)

Answer all questions in the spaces provided.

0 1 Table 1 shows the numbers of breeding pairs of two seabird species on an island before and after a rat eradication programme.

Table 1

Species	Breeding pairs before eradication	Breeding pairs 8 years after eradication
Manx shearwater	45	310
Storm petrel	12	140

0 1 . 1 Explain why oceanic islands have many endemic species but low overall biodiversity.

[4 marks]

0 1 . 2 Use Table 1 to explain why introduced rats had such a severe effect on these birds.

[3 marks]

Do not write outside the box

0 1 . 3 Explain how avian malaria threatens Hawaiian birds and how climate change worsens the threat.

[2 marks]

.....

.....

.....

.....

0 1 . 4 Give **one** other conservation measure for oceanic islands besides eradicating introduced species.

[1 mark]

.....

.....

10

0 2 . 1 Describe the ecological features of mangroves and the adaptations of mangrove trees.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

0 2 . 2 Explain **four** reasons why mangroves are important.

[4 marks]

Do not write outside the box

1 .....

.....

2 .....

.....

3 .....

.....

4 .....

.....

0 2 . 3 Describe the main threats to mangroves and the conservation efforts used.

[3 marks]

.....

.....

.....

.....

.....

.....

.....

10

0 3 Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 3 . 1 Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

.....

.....

Do not write outside the box

Do not write outside the box

0 3 . 2 Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

0 3 . 3 Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

0 3 . 4 Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

10

0 4 . 1 Describe how each of the following threatens Antarctica.

[4 marks]

Global climate change .....

.....

Ozone depletion .....

.....

Tourism .....

.....

Future mineral exploitation .....

.....

0 4 . 2 Describe the main features of the Antarctic Treaty (1959) and the regulations that protect Antarctica.

[6 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

10

Do not write outside the box

0 5 . 1 Which of the following habitats has low biodiversity but a high proportion of endemic species? Tick one box.

[1 mark]

A Tropical rainforest ☐

B Oceanic islands ☐

C Temperate broadleaf woodland ☐

D Mangroves ☐

0 5 . 2 Compare the ecological features of tropical rainforest and Antarctica that explain their very different biodiversity.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

5

0 6 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

.....

.....

.....

.....

.....

.....

.....

.....

Do not write outside the box

0 6 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

0 6 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

Do not write outside the box

Do not write outside the box

Extra space

15

END OF QUESTIONS